Paper 71 — SAM Operates Far from Home

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Eigenvalue regularization as a dynamical phenomenon

Ref 101 | D / G / E; R / B | Must read | Original publication

The paper in one minute

Question: Why does sharpness-aware minimization (SAM) suppress large Hessian eigenvalues *during training*, not only at a final minimum?
Model/Method: Analyze unnormalized SAM in quadratic regression; study stability of its discrete update and compare predicted sharpness with neural-network experiments.
Finding: SAM has a modified edge-of-stability condition, $\eta\lambda(1+\rho\lambda) \simeq 2$, which can stabilize curvature below the GD threshold.
Why we care: SAM's sharpness effect is partly a training-dynamics phenomenon; analyzing only final minima misses the mechanism.

1. Start with the update: what does SAM do?

Ordinary gradient descent uses the current gradient $g = \nabla L(\theta)$. SAM first takes a small step *toward higher loss*, then evaluates the gradient at that perturbed point. With a normalized perturbation, $\delta = \rho g/\|g\|$, this gives $\theta^+ = \theta - \eta\nabla L(\theta+\delta)$. The paper's clean stability derivation instead uses *unnormalized* SAM, $\delta = \rho g$. Distinguishing these two variants matters: the simple equation below is exact for the unnormalized quadratic case, not a universal equation for every SAM implementation.

2. Worked example: large curvature becomes less stable

Consider a one-dimensional quadratic loss $L(x) = \frac{1}{2}\lambda x^2$, with curvature $\lambda > 0$. Its gradient is λx, so ordinary GD updates

$$x^+ = (1-\eta\lambda)x.$$

The distance to zero shrinks exactly when $|1-\eta\lambda| < 1$, or $0 < \eta\lambda < 2$. That is the familiar quadratic edge-of-stability boundary. For unnormalized SAM, first perturb x by $\rho\lambda x$. Evaluate the gradient there:

$$g_{\text{SAM}} = \lambda(x+\rho\lambda x) = (\lambda+\rho\lambda^2)x.$$

The resulting update is therefore

$$\boxed{x^+ = \left[1-\eta(\lambda+\rho\lambda^2)\right]x.}$$

SAM is stable on this quadratic exactly if

$$0 < \eta(\lambda+\rho\lambda^2) < 2.$$

Read the equation, not just the symbols
SAM replaces effective curvature λ by $\lambda+\rho\lambda^2$. A large Hessian eigenvalue therefore hits the stability boundary *earlier* than under ordinary GD. As training adapts, this can create pressure toward lower sharpness.

3. A concrete edge-of-stability calculation

Set learning rate $\eta = 0.5$ and unnormalized SAM radius $\rho = 0.2$. Under GD, the critical curvature is $\lambda_{\text{GD}} = 2/0.5 = 4$. SAM instead reaches its boundary when

$$0.5(\lambda + 0.2\lambda^2) = 2 \quad \Longleftrightarrow \quad 0.2\lambda^2 + \lambda - 4 = 0.$$

The positive root is

$$\lambda_{\text{SAM}} = \frac{-1 + \sqrt{4.2}}{0.4} \approx 2.62.$$

At $\lambda = 3$, ordinary GD has multiplier $1 - 0.5(3) = -0.5$ (stable oscillation), while SAM's multiplier is $1 - 0.5(3 + 1.8) = -1.4$ (unstable on this fixed quadratic). Thus SAM is selective against high curvature at a level GD could still tolerate. This calculation is a *pedagogical quadratic example*, not a neural-network convergence theorem.

4. What the paper actually analyzes

The authors use a quadratic regression model $f(\theta) = y + G^\top\theta + \frac{1}{2}Q(\theta,\theta)$. Its prediction residuals and Jacobian evolve together: a changing model representation changes effective curvature. The paper studies the Jacobian Gram eigenvalues (an NTK-related quantity) and derives the SAM stability condition near a zero-residual solution in Theorem 2.3.

$$\boxed{\eta\lambda_{\max}(1 + \rho\lambda_{\max}) \approx 2.}$$

This is a *local stability boundary*. Its experimentally observed persistence during training is an additional dynamical finding, not a global theorem about all neural-network trajectories.

Why do batch size and normalization matter?

Theorem 2.1 shows weak initial suppression for full-batch training in their simplified random quadratic model. Theorem 2.2 shows minibatch effects can strengthen regularization and introduce dependence on the sampling structure rather than on the full-batch Hessian alone. A normalized SAM perturbation is not identical to the unnormalized calculation above.

Connection to the previous EoS papers
Li et al. (2022) describe progressive sharpening; Chen–Bruna (2023) analyze nonlinear dynamics beyond the usual threshold. Agarwala–Dauphin show that SAM itself shifts the relevant stability boundary.

5. Evidence, contributions, and careful interpretation

Contributions. (i) A dynamical rather than merely final-minimum account of SAM's curvature regularization; (ii) a modified edge-of-stability prediction for unnormalized SAM; (iii) analysis of batch-size effects; and (iv) experiments showing that the modified threshold predicts sharpness in selected networks. **Experimental picture.** In Figure 3, ordinary GD on a fully connected MSE model stabilizes near $\eta\lambda_{\max} = 2$, whereas SAM reaches a smaller leading eigenvalue consistent with $\eta(\lambda_{\max} + \rho\lambda_{\max}^2) \approx 2$. Experiments on WideResNet / CIFAR-10 provide additional evidence that the effect can appear in realistic training.

6. Important assumptions and limitations

- The exact simple threshold is derived for *unnormalized* SAM and a quadratic approximation. Theorem 2.3 concerns a smooth model near a zero-residual solution with nondegenerate Gram spectrum.
- Sharpness stabilization away from minima is an observed phenomenon and supported by simplified-model analysis, not a universal finite-time convergence theorem.
- Batch size can change the dynamics substantially; full-batch reasoning cannot simply be imported into arbitrary minibatch SAM runs.
- A lower Hessian eigenvalue does not by itself prove superior generalization or global optimality.

7. Quick reference

GD stability	$0 < \eta\lambda < 2$ for a positive quadratic eigenmode.
Unnormalized SAM	$\delta = \rho\nabla L$, then $\theta^+ = \theta - \eta\nabla L(\theta + \delta)$.
SAM effective curvature	$\lambda + \rho\lambda^2$ in the fixed-quadratic example.
SAM–EoS	$\eta\lambda_{\max}(1 + \rho\lambda_{\max}) \approx 2$.
Result type	Local stability theorem + empirical dynamical evidence; no universal saddle escape guarantee.

Survey bridge / Papers Master note

SAM's eigenvalue regularization can arise throughout training via a shifted edge-of-stability threshold, not only from penalizing final sharp minima. The exact threshold relies on an unnormalized local quadratic model, while experiments support related dynamics in networks. Useful alongside Foret (2021), Li (2022), and Kalra (2025).